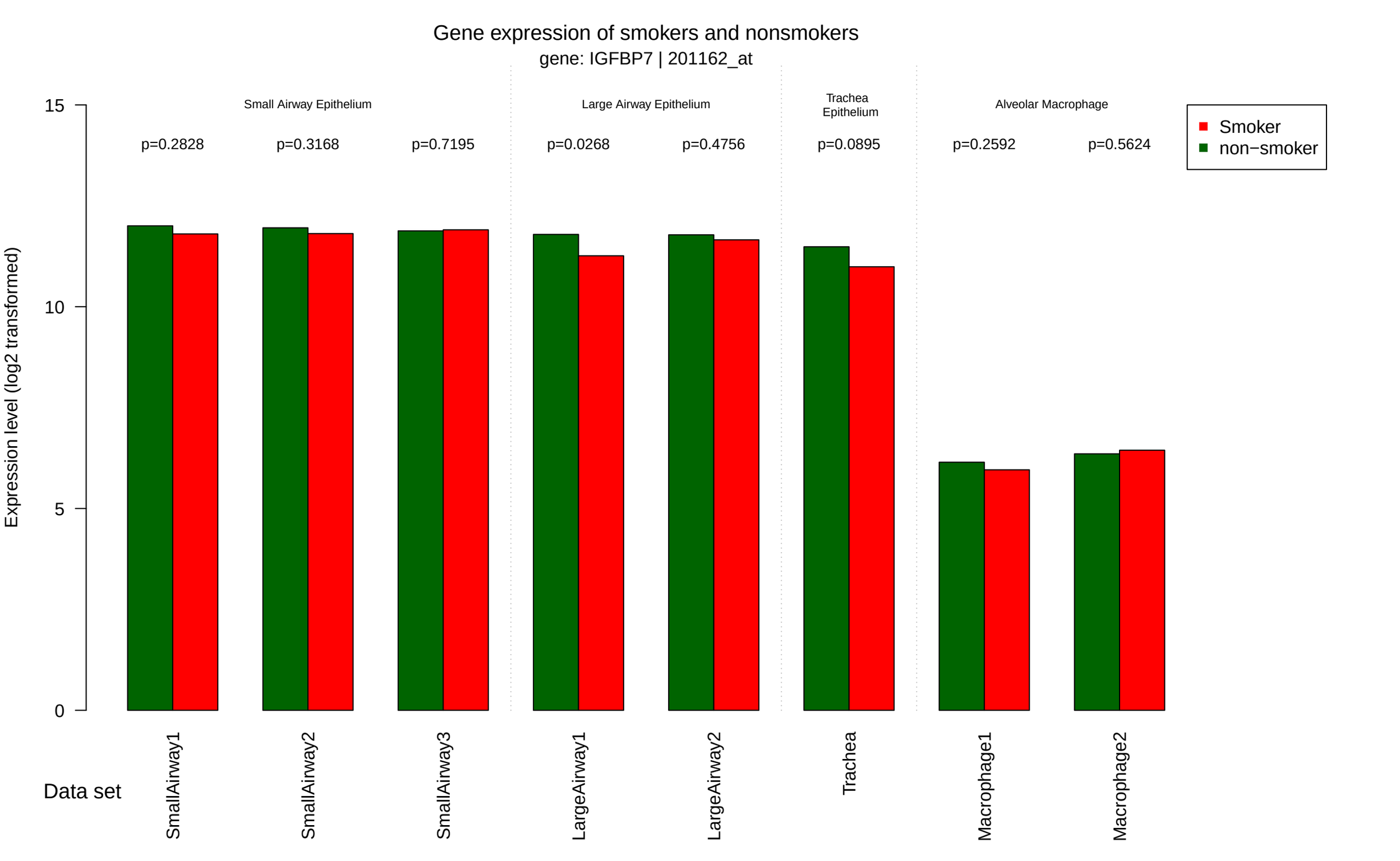
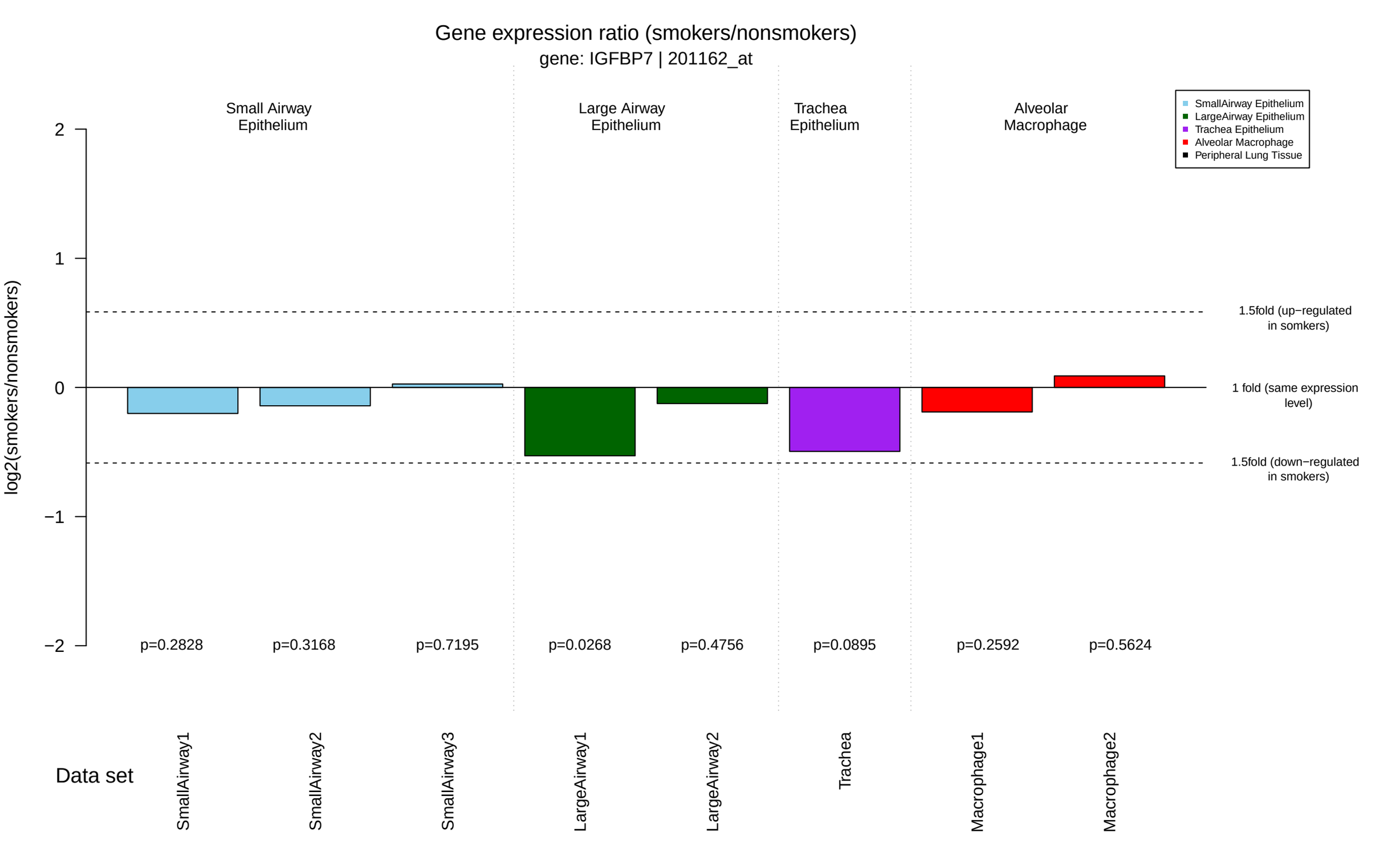
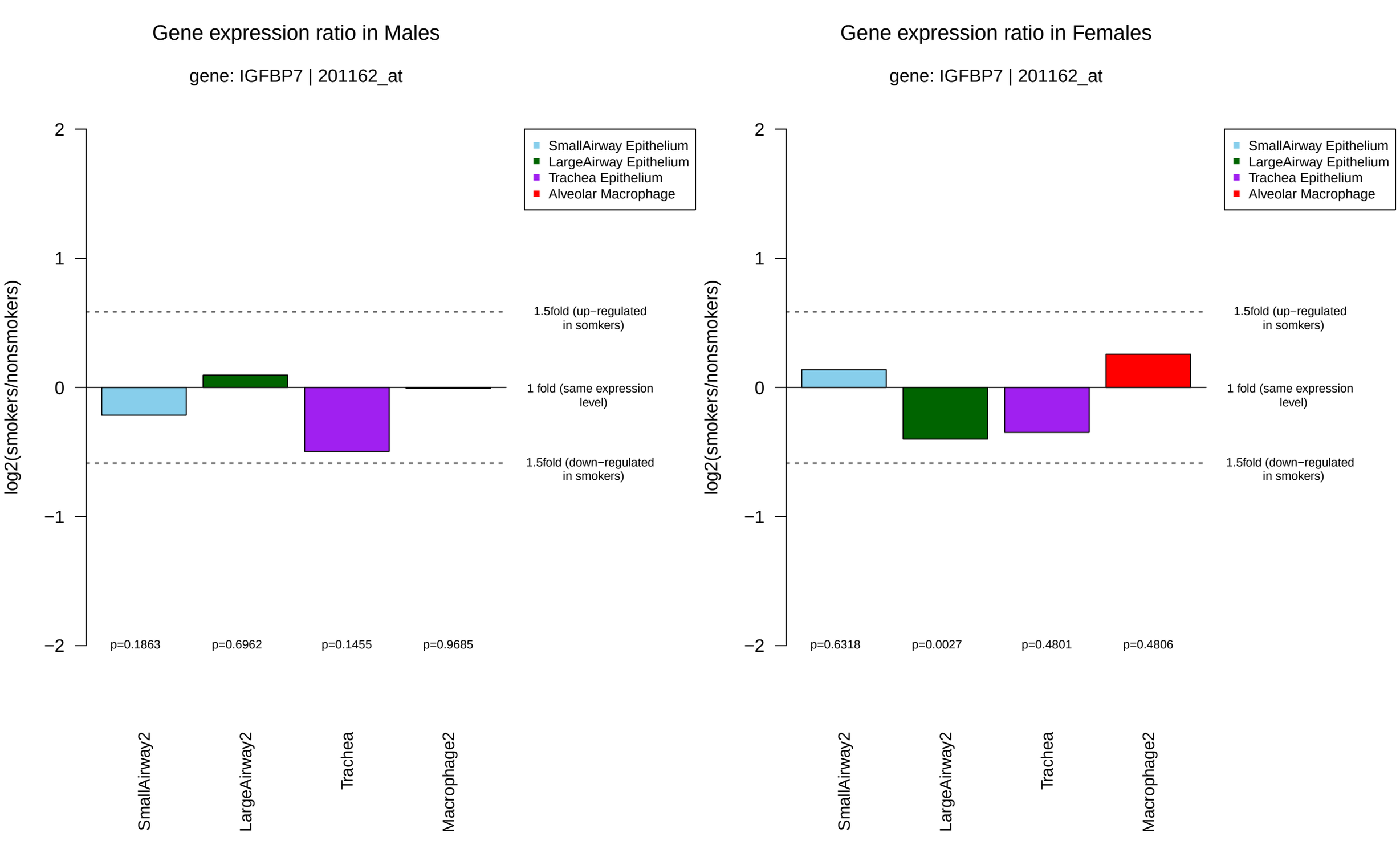
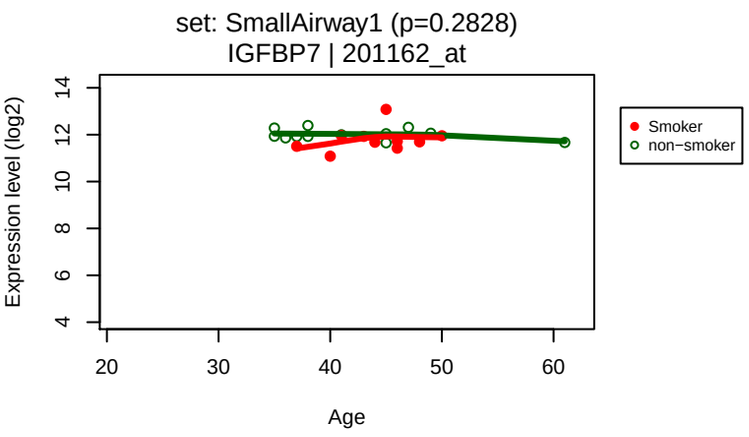
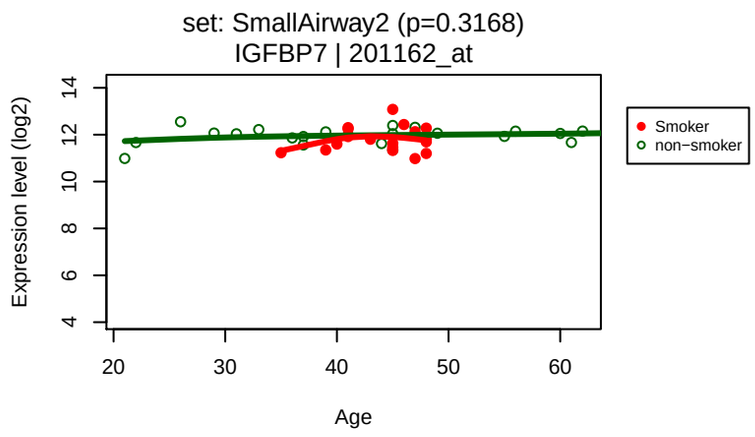
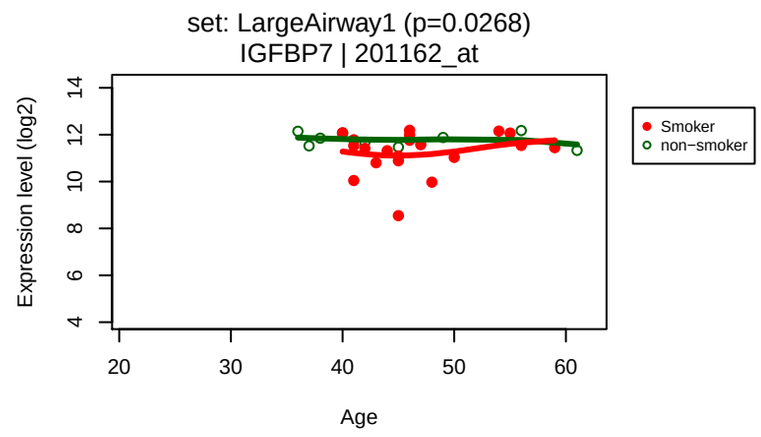

Gene expression of smokers and nonsmokers
gene: IGFBP7 | 201162_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2828
p=0.3168
p=0.7195
p=0.0268
p=0.4756
p=0.0895
p=0.2592
p=0.5624
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

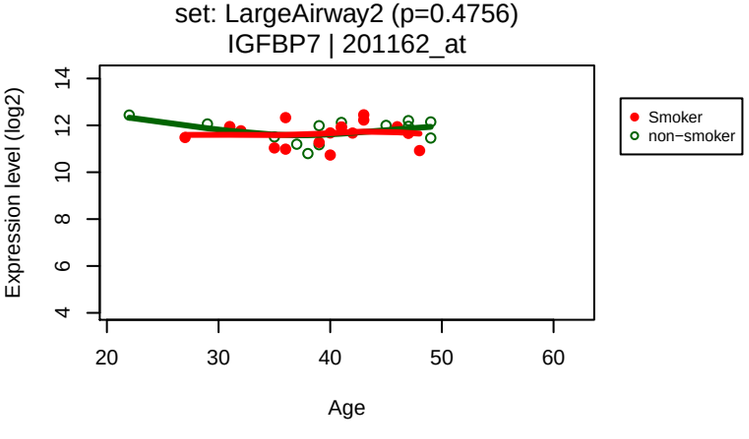

Gene expression ratio (smokers/nonsmokers)
gene: IGFBP7 | 201162_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2828
p=0.3168
p=0.7195
p=0.0268
p=0.4756
p=0.0895
p=0.2592
p=0.5624
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

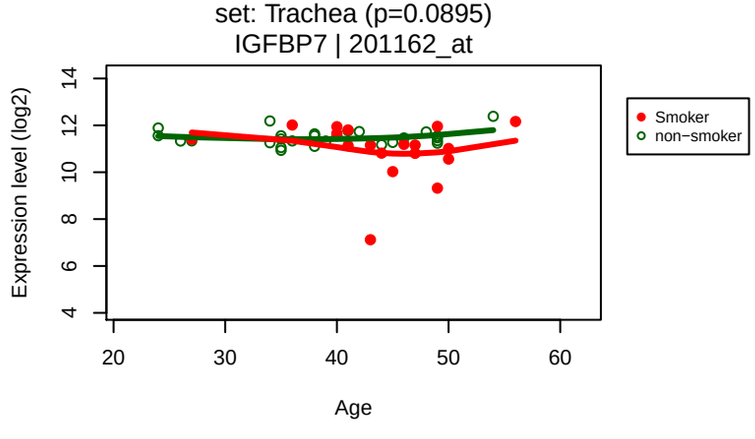

Gene expression ratio in Males
gene: IGFBP7 | 201162_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1863
p=0.6962
p=0.1455
p=0.9685
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IGFBP7 | 201162_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6318
p=0.0027
p=0.4801
p=0.4806
SmallAirway2
LargeAirway2
Trachea
Macrophage2

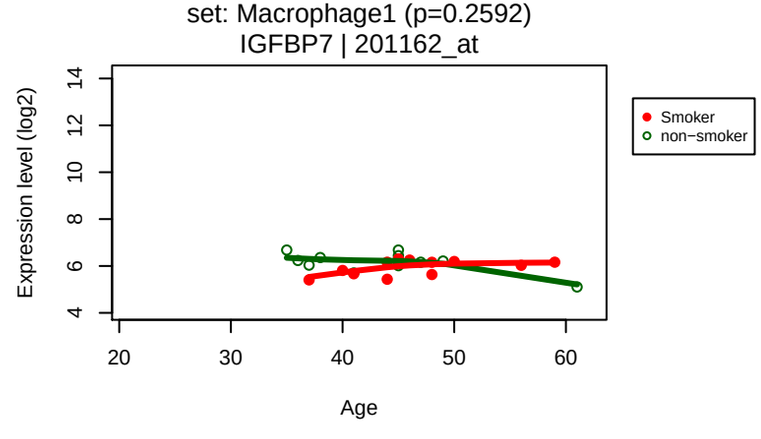

set: SmallAirway1 (p=0.2828)
IGFBP7 | 201162_at
Expression level (log2)
Age
Smoker
non−smoker

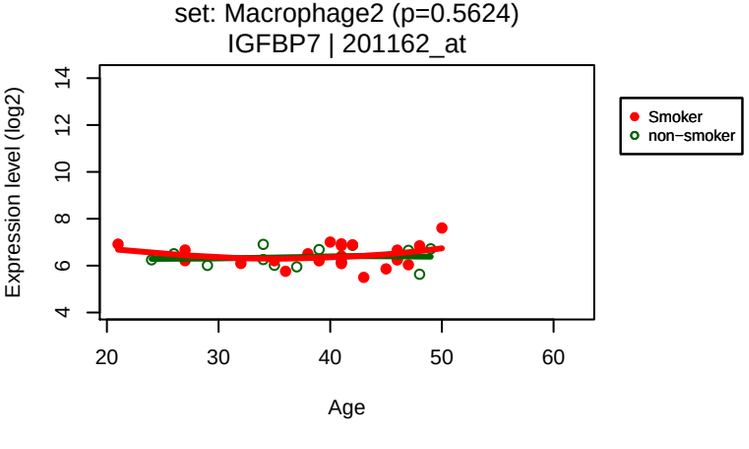

set: SmallAirway2 (p=0.3168)
IGFBP7 | 201162_at
Expression level (log2)
Age
Smoker
non−smoker

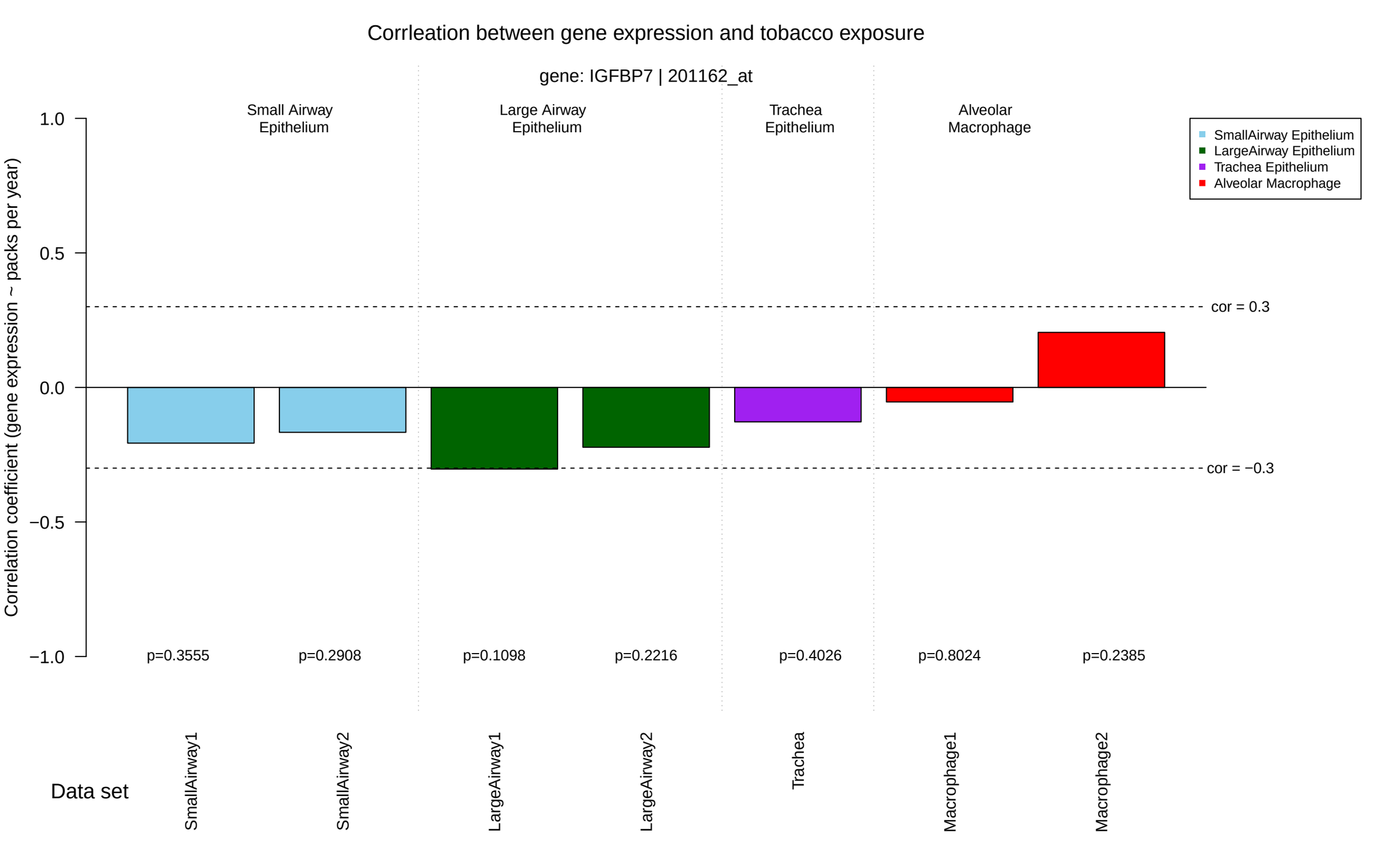

set: LargeAirway1 (p=0.0268)
IGFBP7 | 201162_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4756)
IGFBP7 | 201162_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0895)
IGFBP7 | 201162_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2592)
IGFBP7 | 201162_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5624)
IGFBP7 | 201162_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IGFBP7 | 201162_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3555
p=0.2908
p=0.1098
p=0.2216
p=0.4026
p=0.8024
p=0.2385
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2